

## Fixed width addition

**Fixed-width addition:** adding one bit at time, using the usual column-by-column and carry arithmetic, and dropping the carry from the leftmost column so the result is the same width as the summands. *Does this give the right value for the sum?*

$$\begin{array}{r} [0\ 1\ 0\ 1]_{s,4} \\ + [1\ 1\ 0\ 1]_{s,4} \\ \hline \end{array}$$

$$\begin{array}{r} [0\ 1\ 0\ 1]_{2c,4} \\ + [1\ 0\ 1\ 1]_{2c,4} \\ \hline \end{array}$$

$$\begin{array}{r} (1\ 1\ 0\ 1\ 0\ 0)_{2,6} \\ + (0\ 0\ 0\ 1\ 0\ 1)_{2,6} \\ \hline \end{array}$$

$$\begin{array}{r} [1\ 1\ 0\ 1\ 0\ 0]_{s,6} \\ + [0\ 0\ 0\ 1\ 0\ 1]_{s,6} \\ \hline \end{array}$$

$$\begin{array}{r} [1\ 1\ 0\ 1\ 0\ 0]_{2c,6} \\ + [0\ 0\ 0\ 1\ 0\ 1]_{2c,6} \\ \hline \end{array}$$

# Circuits basics

In a **combinatorial circuit** (also known as a **logic circuit**), we have **logic gates** connected by **wires**. The inputs to the circuits are the values set on the input wires: possible values are 0 (low) or 1 (high). The values flow along the wires from left to right. A wire may be split into two or more wires, indicated with a filled-in circle (representing solder). Values stay the same along a wire. When one or more wires flow into a gate, the output value of that gate is computed from the input values based on the gate's definition table. Outputs of gates may become inputs to other gates.

## Logic gates definitions

| Inputs |     | Output      |
|--------|-----|-------------|
| $x$    | $y$ | $x$ AND $y$ |
| 1      | 1   | 1           |
| 1      | 0   | 0           |
| 0      | 1   | 0           |
| 0      | 0   | 0           |



| Inputs |     | Output      |
|--------|-----|-------------|
| $x$    | $y$ | $x$ XOR $y$ |
| 1      | 1   | 0           |
| 1      | 0   | 1           |
| 0      | 1   | 1           |
| 0      | 0   | 0           |

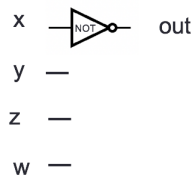


| Input | Output  |
|-------|---------|
| $x$   | NOT $x$ |
| 1     | 0       |
| 0     | 1       |



# Digital circuits basic examples

Example digital circuit:



Output when  $x = 1, y = 0, z = 0, w = 1$  is \_\_\_\_\_

Output when  $x = 1, y = 1, z = 1, w = 1$  is \_\_\_\_\_

Output when  $x = 0, y = 0, z = 0, w = 1$  is \_\_\_\_\_

Draw a logic circuit with inputs  $x$  and  $y$  whose output is always 0. *Can you use exactly 1 gate?*